

Thomas Polstra — Curriculum Vitae

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Employment

- **University of Alabama**
Assistant Professor 2022–present
- Postdoctoral Experience.....
 - **Mathematical Sciences Research Institute**
Offsite postdoctoral fellow 2021–2022
Host Institution: University of Virginia
 - **University of Virginia**
Research associate and lecturer 2020–2021
 - **University of Utah**
NSF postdoctoral fellow 2017–2020

Research Interests

Commutative Algebra and Algebraic Geometry.

Education

- PhD, Mathematics.....
 - **University of Missouri-Columbia**
Advisor: Ian Aberbach 2012–2017
- Bachelors, Mathematics.....
 - **Georgia State University**
Advisor: Florian Enescu 2008–2012

Awards and Funding Information

- External Funding & Awards.....
 - National Science Foundation standard grant DMS #2502317, \$278,488, 2025–Present.
 - National Science Foundation standard grant DMS #2101890, \$105,000, 2021–2024.
 - National Science Foundation conference grants
 - Morgantown Algebra Days 2019, Morgantown, WV.
 - Commutative Algebra in the South 2023, Tuscaloosa, AL.
 - Simons postdoctoral fellowship, 2021–2022.
 - National Science Foundation postdoctoral fellowship, 2017–2020.
- Internal Funding & Awards.....
 - University of Alabama, President's Faculty Research Award, Spring 2026.

- University of Alabama, Barefield College of Arts and Sciences ASPIRE program, Fall 2025.
- University of Alabama Department of Mathematics Focused Research Group, 2024.
- Outstanding postdoc award, University of Utah, Spring 2019.

Publications and Preprints

1. Bertini Theorem for F -rationality is false, with Austyn Simpson, submitted.
2. Bertini Theorems for F -rational and F -pure singularities, with Alessandro De Stefani and Austyn Simpson, submitted.
3. F -injectivity does not imply F -fullness in normal domains, with Alessandro De Stefani and Austyn Simpson, submitted.
4. Strong F -regularity and the Uniform Symbolic Topology Property, submitted.
5. Zariski-Nagata Theorems for Singularities and the Uniform Izumi-Rees Property, *Compositio Mathematica*. 2026;162(3):602-629.
6. On F -pure Inversion of Adjunction, with Austyn Simpson and Kevin Tucker, Higher dimensional algebraic geometry—a volume in honor of V. V. Shokurov, London Math. Soc. Lecture Note Ser., vol. 489, Cambridge Univ. Press, Cambridge, 2025, pp. 319–344.
7. On the Equality of Test Ideals, with Ian Aberbach and Craig Huneke, *Adv. Math.* 441, (2024).
8. Primary Decompositions of Regular Sequences, submitted.
9. On the Applications of Big Cohen-Macaulay Modules and Algebras, AMS Contemporary Mathematics, Lectures on Tight Closure and Its Applications, December, 2025.
10. F -singularities: a commutative algebra approach, with Linquan Ma, preprint available.
11. Compatible ideals in $\mathbb{Q}$ -Gorenstein rings, with Karl Schwede, *Proc. Amer. Math. Soc.* 151 (2023), 4099-4112.
12. Local Cohomology Bounds and the Weak Implies Strong Conjecture in Dimension 4, with Ian Aberbach, *Journal of Algebra* Volume 605, (2022), 37–57.
13. F -purity deforms in $\mathbb{Q}$ -Gorenstein rings, with Austyn Simpson, *Int. Math. Res. Not. IMRN* 2023, no. 24, 20725–20747.
14. Global F -splitting ratio of modules, with De Stefani, Alessandro De Stefani and Yongwei Yao, *Journal of Algebra* Volume 610, (2022), 773–792.
15. A theorem about maximal Cohen-Macaulay modules, *International Mathematics Research Notices*, Volume 2022, Issue 3, February 2022, Pages 2086–2094.
16. Global Frobenius Betti numbers and Frobenius Euler characteristic, with Alessandro De Stefani and Yongwei Yao, *Michigan Math. J. Advance Publication* 1 - 20, 2021.
17. F -nilpotent rings and permanence properties, with Jenny Kenkel, Kyle Maddox, and Austyn Simpson, *J. Commut. Algebra* 15 (2023), no. 4, 559–575.
18. Equimultiplicity theory of strongly F -regular rings, with Ilya Smirnov, *Michigan Math. J.* 70(4): 837-856 (October 2021).
19. Coverings of rational double points in mixed characteristic, with Javier Carvajal-Rojas, Linquan Ma, Karl Schwede, and Kevin Tucker, *Journal of Singularities*, 23 (2021), 127-150.
20. Globalizing F -invariants, with Alessandro De Stefani and Yongwei Yao, *Adv. Math.* 350 (2019), 359–395.

21. F -signature under birational morphisms, with Linquan Ma, Karl Schwede, and Kevin Tucker, Forum Math. Sigma 7 (2019), e11, 20 pp.
22. Nilpotence of Frobenius actions on local cohomology modules and Frobenius closure of ideals, with Pham Hung Quy, J. Algebra 529 (2019), 196–225.
23. Continuity of Hilbert-Kunz multiplicity and F -signature, with Ilya Smirnov, Nagoya Mathematical Journal, 1-24. doi:10.1017/nmj.2018.43.
24. F -signature and Hilbert-Kunz multiplicity: A combined approach and comparison, with Kevin Tucker, Algebra Number Theory 12 (2018), no. 1, 61–97.
25. A generalization of Serre's Theorem and Bass's Cancellation Theorem on projective modules, with Alessandro De Stefani and Yongwei Yao, Proc. Amer. Math. Soc. 146 (2018), no. 4, 1417–1430.
26. Uniform bounds in F -finite rings and lower semi-continuity of the F -signature, Trans. Amer. Math. Soc. 370 (2018), no. 5, 3147–3169.
27. Depths and Stanley depths of path ideals of spines, with Daniel Campos, Ryan Gunderson, Susan Morey, Chelsey Paulsen, Involve 9 (2016), no. 1, 155–170.
28. Depths and Cohen-Macaulay properties of path ideals with Daniel Campos, Ryan Gunderson, Susan Morey, Chelsey Paulsen, J. Pure Appl. Algebra 218 (2014), no. 8, 1537–1543.
29. Local cohomology bounds and test ideals, with Ian Aberbach, permanent preprint due to delays in publication.